Problem Set: INDE301_Ch1_Notes_24.pdf

Generated using AI-powered agent orchestration system.

Chapter Analysis

Topics Covered: - Engineering Economy - Decision-Making Process - Economic Analysis in Engineering

Key Formulas:

No formulas identified

Problem 1

Difficulty: EASY

Topic: Decision-Making Process

Problem Statement

List the first three steps in the decision-making process as outlined in the chapter.

Find: - List the first three steps in the decision-making process.

Problem 2

Difficulty: MEDIUM

Topic: Estimating financial outcomes

Problem Statement

A company is considering two alternatives for a new project. Alternative A has an estimated cost of $150,000 and is expected to generate $200,000 in revenue. Alternative B has an estimated cost of $120,000 with expected revenue of $180,000. Calculate the net financial outcome for each alternative.

Given: - Cost of Alternative A: $150,000 - Revenue from Alternative A: $200,000 - Cost of Alternative B: $120,000 - Revenue from Alternative B: $180,000

Find: - Net financial outcome for Alternative A - Net financial outcome for Alternative B

Problem 3

Difficulty: HARD

Topic: Evaluating costs

Problem Statement

A company has three alternatives for upgrading its manufacturing process. The estimated costs and expected savings over five years are as follows: Alternative X costs $500,000 and saves $600,000; Alternative Y costs $450,000 and saves $550,000; Alternative Z costs $400,000 and saves $450,000. Evaluate each alternative based on the net savings and determine which alternative is the best choice.

Given: - Cost of Alternative X: $500,000, Savings: $600,000 - Cost of Alternative Y: $450,000, Savings: $550,000 - Cost of Alternative Z: $400,000, Savings: $450,000

Find: - Net savings for Alternative X - Net savings for Alternative Y - Net savings for Alternative Z - Best alternative based on net savings